Code No: 118BJ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2020
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Briefly describe the distinction between requirements and specification.
b) Explain the formalisms for system design. [7+8]

2.a) Explain the memory organization of 8051 in detail.
b) Explain the operation of counters and timers in 8051. [5+10]

3.a) Explain the assembly language programming process.
b) Explain the Data Transfer Instructions in 8051 with examples. [5+10]

4.a) What are the different logical instructions are available in 8051? Write their syntax.
b) Write an assembly language program to 8-bit addition and subtraction. [7+8]

5.a) Explain about serial data communication modes in 8051.
b) Draw and explain the A/D and D/A converters. [7+8]

6.a) Explain task scheduling mechanisms with an example.
b) Explain how interrupt routines handled in RTOS Environment. [8+7]

7.a) Discuss Software development tools of embedded system.
b) Write short notes on the following
i) Testing on host machine
ii) Laboratory tools. [7+8]

8.a) Discuss briefly about SHARC architecture.
b) Write about Internet enabled systems. [9+6]

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NTUH Used Papers 20-09-2020